

Where every new part goes

Four views of the stock coach, with the new hardware placed in it: the floor plan, a side section, the underbody and the roof. Each new part has a number that matches the schedule below, which says where it goes, what it takes the place of, how it mounts and what it connects to. Positions are inches forward of the rear wall. They come from the model and get confirmed with a tape measure.

INSIDE THE CABIN · 3 CHANGES - RV5 hub, disconnect and PV breakers go in the factory desk base, roadside rear corner under the lift bed (1, 3) - Epad on the wall at the desk (4) - Transfer switch and surge protector in the pantry base (11, 12) - Washer cabinet at A or B, both under the lift-bed area (13)	ROOF - Six panels in one crosswise row, split around the AC (6) - Combiner and cable gland (7, 5) - Chill Cube swaps into the existing opening (8) - Starlink Mini beside it (9)	UNDERNEATH AND CAB - Sealed box with both packs under the tail (2) - Two Orions under the passenger swivel seat (10)	STILL OPEN - Washer: A (bath side) or B (rear garage) - Six fit checks at the walkthrough, listed below
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Bluetti / 48 V DC

Solar

Alternator charging

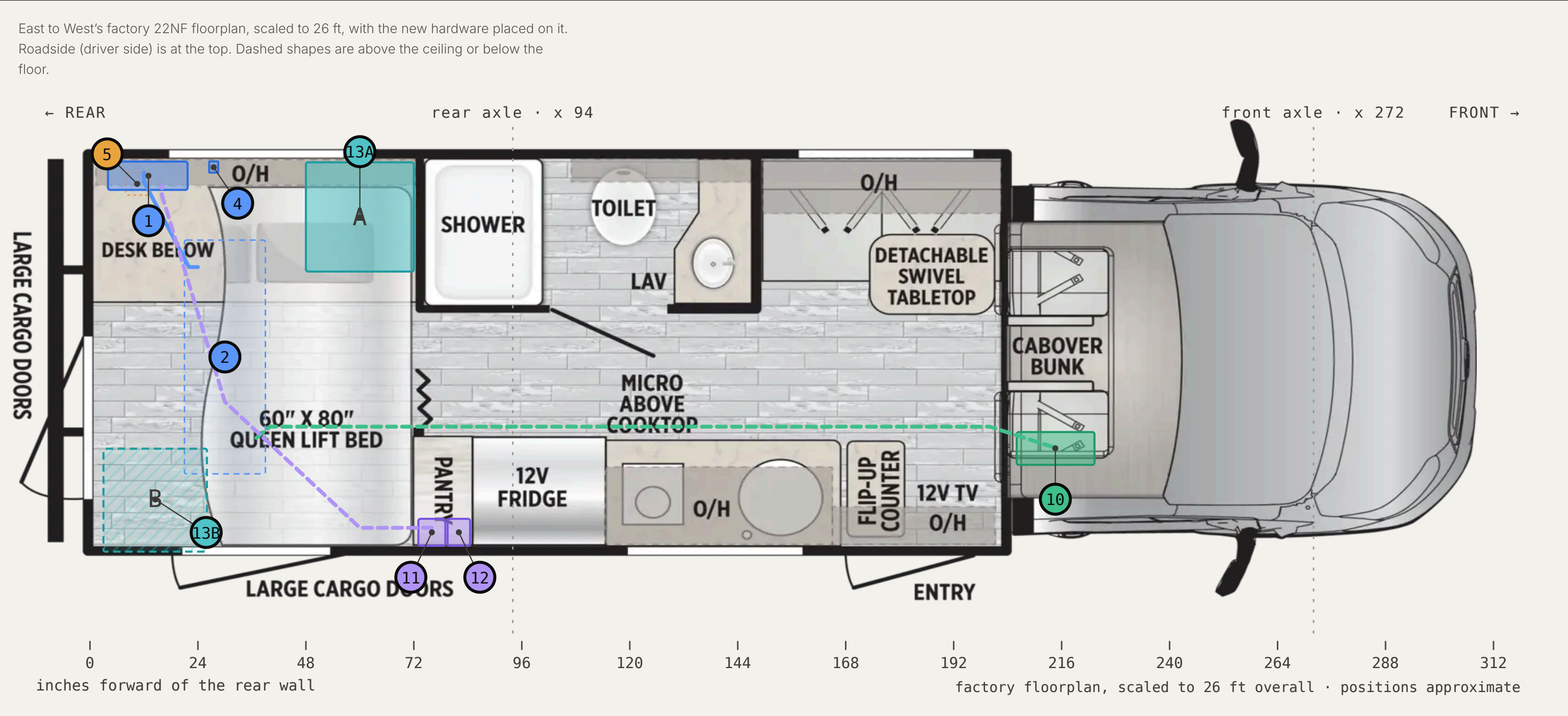
120 V AC

Washer-dryer

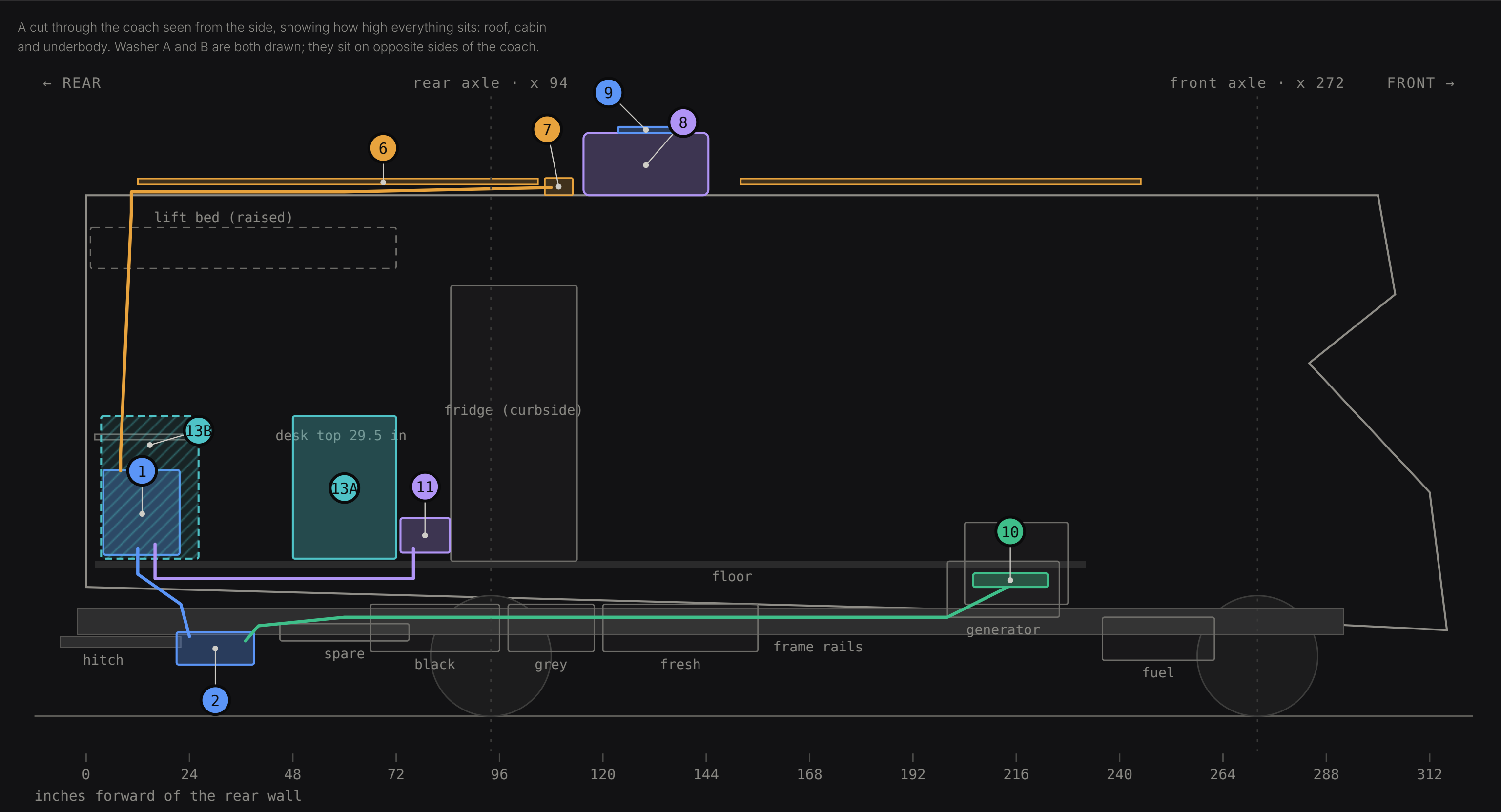
Stock coach

Fit check

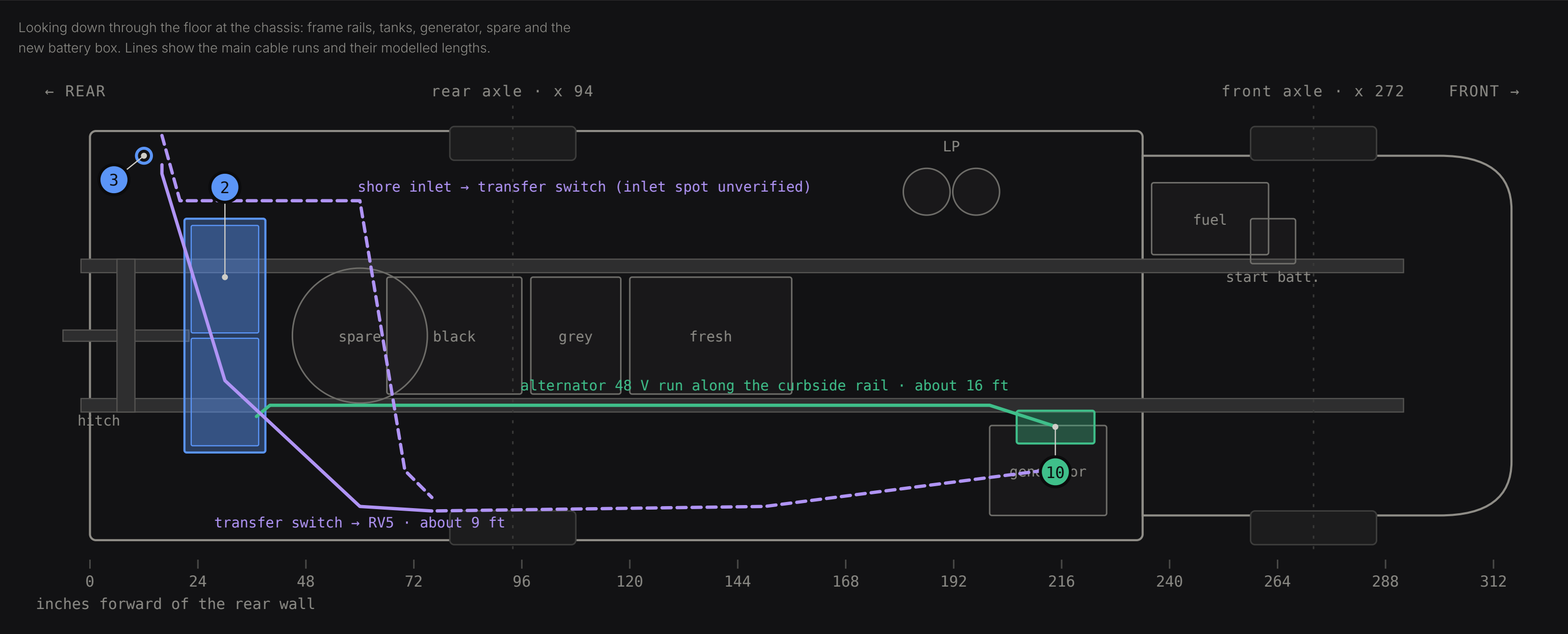
Floor plan



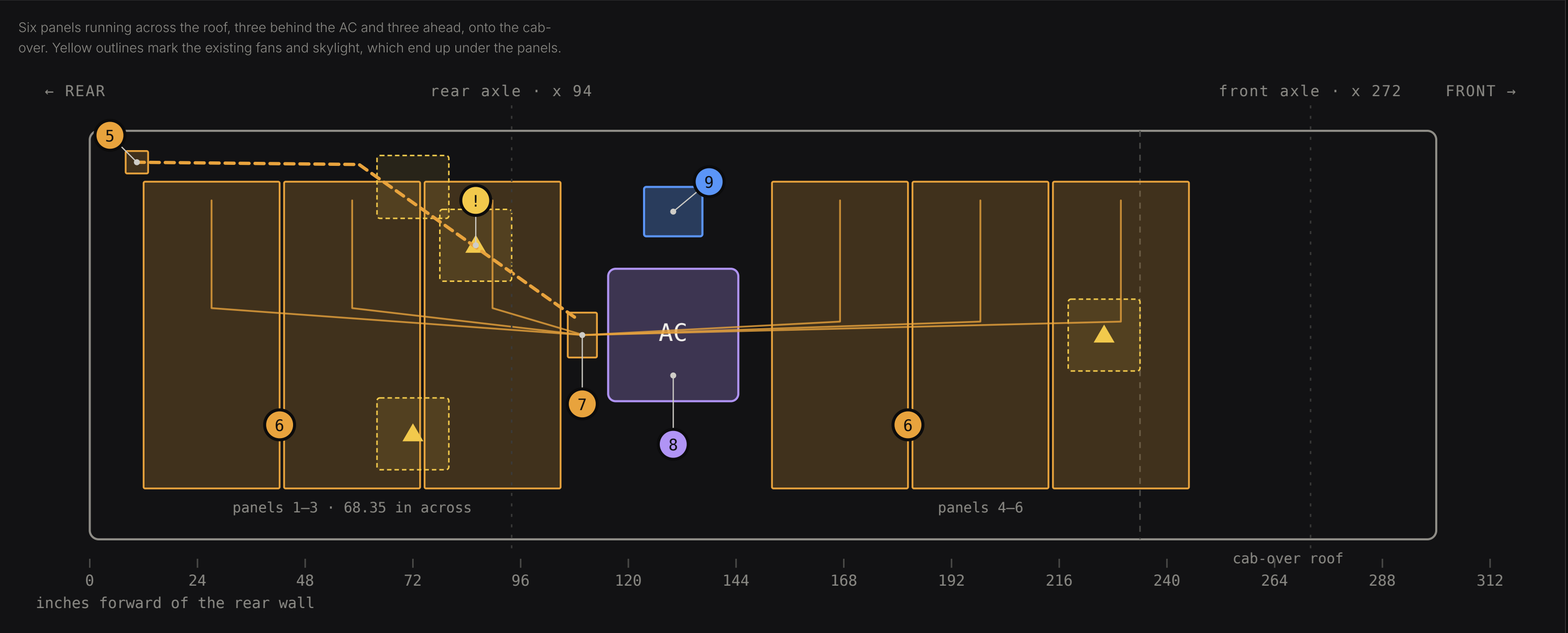
Side section



Underbody



Roof



PLACEMENT SCHEDULE

Every new part, where it goes and what it connects to

#	PART · WHERE IT GOES	POSITION	TAKES THE PLACE OF	MOUNT AND ACCESS	CONNECTS TO
1	Bluetti RV5 hub In the base of the factory desk ("desk below"), roadside rear corner under the lift bed, on the floor against the roadside wall.	x 4–22 in from rear wall · roadside wall · floor level	Storage space in the desk base.	Bolted to the floor through its mounting holes. Pedestal needs a vented front panel so the RV5 keeps 7.87 in of open air at its vents. Front panel removable for service.	48 V down through the floor to the packs (3 ft) · PV trunks from the roof gland straight above (14 ft each) · 120 V in from the transfer switch in the pantry base (about 9 ft) · 120 V out to the factory breaker panel (location to find) · 12 V out to the coach 12 V panel.
2	2 × B4810 in a sealed box Under the tail, across the frame rails, between the hitch crossmember and the spare tire.	x 21–39 in from rear wall (about 64 in behind the rear axle) · 52 × 18 × 7.5 in box · bottom about 12 in off the ground	Nothing inside. Uses empty space behind the spare. Must clear the hitch receiver tube, the exhaust tip, and the dump valves.	Hangers bolted to the coach builder's frame extension (not welded). Lockable lid, baffled vent, low drain, skid plate. Packs fused inside the box at their terminals.	48 V up through the floor into the desk base (3 ft) · alternator 48 V run from the cab (about 16 ft).

#	PART · WHERE IT GOES	POSITION	TAKES THE PLACE OF	MOUNT AND ACCESS	CONNECTS TO
3	Main DC disconnect, PV breakers, pack fuse Disconnect and both PV breakers inside the desk base next to the RV5. The pack fuse goes in the battery box.	Desk base, beside #1 · pack fuse in #2	Shares the desk base with the RV5.	Reachable from inside the coach without crawling under. (The 3D model draws the disconnect on the box; this map moves it inside.)	Sits between the packs and the RV5, and between each PV trunk and its RV5 input.
4	Bluetti Epad (control screen) On the roadside wall at the desk, just above the desk base.	x 26–28 in · roadside wall · eye level when seated	Wall space only.	Surface mount. Ethernet to the RV5 directly below.	Data cable to #1 (under 3 ft).
5	Roof cable gland Rear roadside corner of the roof, directly above the desk.	x 8–13 in · 36–41 in roadside of centre	Nothing. New sealed penetration (or the factory solar side-port if it lands here).	Sealed gland with written roof-warranty coverage. The two PV trunks drop inside the rear roadside wall into the desk base.	Both PV trunks from #7 to #1.
6	6 × Callsun 275 W panels Roof, one crosswise row of six: panels 1–3 behind the AC, panels 4–6 ahead of it, running onto the cab-over roof.	x 12–105 and 152–245 in · each 68.35 in across the roof	Roof space only. The existing fans and skylight sit under the panels (see fit check).	Z-brackets or a low rail, about 2.5 in off the roof. Walk margin left on both sides.	Panel leads to #7 (4–12 ft each, about 48 ft total).
7	Combiner box (3 + 3, 25 A per panel) Roof centreline, between panels 1–3 and the AC.	x 106–113 in · centreline	Nothing.	Bolted to the roof, sealed. Two groups of three, each with a 25 A fuse per panel.	Panels in; two PV trunks out to #5 then #1 (14 ft each).
8	Furrion Chill Cube 18K (ducted) The existing 14 × 14 in AC opening. Swaps one-for-one with the factory unit.	Opening modelled at x 116–145 in (centre x 130). Real position unverified.	The factory AC (removed). Its ceiling assembly is replaced by the Chill Cube air distribution box, which feeds the existing ducts.	Bolts through the existing opening. Confirm the air distribution box mates to the factory ducts before buying.	120 V from the main panel (RV5 output).
9	Starlink Mini Roof, roadside of the AC on a flat mount.	x 124–137 in · 22–33 in roadside of centre	Roof space only.	Flat mount, adhesive or bolted. Cable rides the PV gland or the AC chase.	12 V from the RV5 12 V output.
10	2 × Victron Orion-Tr 12/48 Under the passenger (curbside) cab seat, close to the starter battery.	x 206–223 in · curbside · under the seat base	Space under the passenger seat. The cab seats swivel to face the table, so the swivel base must leave room (fit check).	Screwed to the seat base or the floor beside it. Ignition-switched. Fuse at the starter battery.	12 V from the starter battery (about 3 ft) · 48 V back to the packs along the curbside rail (about 16 ft, about 16 A, so a small cable).
11	Transfer switch In the base of the pantry, curbside, just forward of the lift bed. Candidate spot: move it next to the factory power center if that turns out to be nearby.	x 73–79 in · curbside · floor of the pantry	The bottom shelf of the pantry.	Screwed to the floor. Service through the pantry door.	Generator in (length depends on the generator bay, unverified) · shore in through #12 · out to the RV5 AC input (about 9 ft).
12	Surge protector / EMS (hardwired) Beside the transfer switch in the pantry base.	x 79–85 in · curbside	The bottom shelf of the pantry.	Screwed to the floor next to #11.	Shore inlet in (inlet location unverified) · out to #11.
13A	Splendide WDV2200XCD, location A Rear office, roadside: replaces the bath-side drawer stack at the forward end of the lift-bed area, backed against the shower wall (owner's October 5 plan).	x 48–72 in · roadside, behind the shower	The bath-side drawer stack under the lift bed.	Own cabinet, kept separate from the desk. Vent straight out the roadside wall. Water from the shower plumbing on the other side of the wall; drain to the grey tank.	Own 15 A circuit from the main panel.
13B	Splendide WDV2200XCD, location B Rear garage: curbside rear corner under the lift bed, across from the desk.	x 3–26 in · curbside rear corner	Garage floor space on the curbside.	Vent out the rear wall. Sits right inside the curbside large cargo door, so it blocks that door and part of the garage. Longer water and drain runs than A, and more rear-axle load.	Own 15 A circuit from the main panel.

Cable lengths are measured along the routes drawn in the 3D model, rounded to the nearest foot.

ROOF FIT

What the six-panel row needs, and what to do if it doesn't fit

LENGTH THE ROW NEEDS	ROOF THE MODEL USED	MAXXFAN UNDER A PANEL
233 in	~300 in, estimate	No
Six panels at 30.3 in, 1 in gaps, 10.5 in for the combiner before the AC, 29 in of AC, and 7.5 in after it. In the model it runs from x 12 to x 245.	The model treats the roof as flat from the rear cap over the cab-over. The row starts 12 in from the rear cap and ends 11 in past the house-to-cab-over seam (x 234). The 1 in margin quoted earlier compared the row with the house section alone. Measure both sections.	A MaxxFan Deluxe with rain dome stands 5.0 in tall closed and 9.1 in open. Panels on 2.5 in brackets can't pass over it.

FALLBACKS, IN ORDER

1. **Tall rails over the fans.** Raise the panels so their underside sits about 10 in off the roof, clearing an open MaxxFan. Panel tops end up about 11.4 in high, still under the Chill Cube's 14.5 in, so the coach's overall height doesn't change. The cost is a heavier rack and more wind load.
2. **Leave a fan open.** Shift the row so a fan falls in a gap. Each uncovered fan costs one panel slot, leaving five panels: 1,375 W wired 3 + 2. Two panels in parallel need no per-panel fuses.
3. **Short roof.** Drop the front panel: five panels need about 202 in. Four panels (2 + 2, 1,100 W) need about 171 in.
4. **Tighten the gaps.** Move the combiner off the row to the roof edge to save about 6 in.

REAR AXLE CHECK

Axle loads added by the build, against the Transit 350 ratings

The coach's empty axle weights aren't published, so the table works backwards. It totals everything the build and a loaded trip add to each axle, then shows the heaviest the *empty* axle can weigh and still stay under its rating (front 4,630 lb, rear 7,275 lb). Included: the build at its modelled positions, a full fresh tank (290.5 lb at the tank), a 40 lb tow bar on the receiver, a 200 lb driver, and 350 lb of gear spread midship.

SCENARIO	ADDED TO FRONT AXLE	ADDED TO REAR AXLE	EMPTY REAR AXLE CAN BE AT MOST	EMPTY FRONT AXLE CAN BE AT MOST
Washer A · panels 29.8 lb each	+237 lb	+1342 lb	5,933 lb	4,393 lb
Washer A · panels 65.7 lb each	+278 lb	+1516 lb	5,759 lb	4,352 lb
Washer B · panels 29.8 lb each	+199 lb	+1381 lb	5,894 lb	4,431 lb
Washer B · panels 65.7 lb each	+240 lb	+1554 lb	5,721 lb	4,390 lb

- **What it means.** With an empty coach around 9,000 lb, the rear axle stays legal if the empty rear axle weighs no more than about 5,550–5,990 lb, or roughly 62–67% of the empty weight on the rear. The front axle has room in every case. A CAT scale reading of the empty coach answers it.
- **Gear placement matters as much as the box.** Putting the 350 lb of gear in the rear garage instead of midship adds another 177 lb to the rear axle. Every 100 lb moved from the garage to midship takes about 50 lb off the rear.
- **Should the box move forward?** Moving the packs from 64 in behind the axle to directly over it takes 82 lb off the rear axle. In the model, the space between the rails at the axle is taken by the tanks. Keep the tail

box unless the scale shows the empty rear axle above about 5,500 lb. If it does, move gear forward first, then look at the box.

BATTERY CABLE

The 48 V run from the packs to the RV5 is about 3 ft

The box sits under the rear floor, and the RV5 sits in the desk pedestal right above it and slightly behind. The modelled run goes down through the pedestal floor and into the box: 37 in, about 3 ft. The 10–15 ft guess came from earlier layouts with the hub further forward. The longest 48 V run is now the alternator line from the Orions: about 20 ft at about 16 A, so a small cable. The pack fuse stays at the pack terminals inside the box.

FIT CHECKS

Nine things the tape measure has to confirm

1. **Roof fans and skylight under the panels.** The panels sit 2.5 in off the roof, and the model puts two roof fans and the bath skylight under them. A MaxxFan Deluxe with rain dome is 23.2 × 16.6 in and stands 5.0 in tall closed and 9.1 in with the lid open (retailer spec, overall unit height). It cannot sit under a panel on 2.5 in brackets. See the roof fallback section.
2. **Roof length and AC position.** The row of six spans 233 in from the first panel edge to the last (x 12 to 245). The model treats the roof as flat from the rear cap over the cab-over to about x 300. Measure the flat roof length, whether the cab-over roof is flat and walkable, and the AC opening's distance from the rear cap.
3. **Tail box vs. hitch and spare.** In the model the box starts right where the receiver tube ends. Measure the gap from the hitch crossmember to the spare, frame-extension height and bolt points, and exhaust-tip position.
4. **Desk base size.** The RV5 is 17.7 × 19.7 × 6.3 in and wants 7.87 in of air at its vents. Measure the factory desk base in the roadside rear corner to confirm it can take the RV5 with a vented front.
5. **Passenger seat swivel.** The cab seats swivel to face the table. Check what's under the passenger swivel base and whether two Orions (each about 7.3 × 5.1 × 3.1 in) fit without blocking the swivel. Fallback: the floor beside the seat base or the pantry base with the transfer switch.
6. **Factory power center and generator bay.** Find the factory breaker panel and converter, and the generator bay. The transfer switch goes in the pantry base unless the power center is close by.
7. **Shore inlet location.** The model guesses the inlet is at the rear roadside. If it's forward, the shore run to the transfer switch gets shorter.
8. **Cargo doors.** The factory floorplan shows large cargo doors in the rear wall and on the curbside under the bed. Washer B sits inside the curbside door. The RV5 and desk are in the roadside rear corner, clear of the curbside door but next to the rear doors.
9. **Washer A space.** Location A needs the bath-side drawer stack under the lift bed to be at least 24 in wide, with clear height under the bed for the 33.1 in machine plus its top.

3D REFERENCE

The same layout in the rendered model

